POC · GENERATIVE UI

GenUI Studio

One AI agent that builds web pages from a fixed library of React components, and edits them when you ask.

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Vercel AI SDK · json-render · Next.js

THE ASK

Three questions about generative UI

1

One agent, many page layouts

Can a single agent produce very different page structures?

2

Components from small components

How are bigger UI pieces composed from small building blocks?

3

Following instructions

How accurately does the agent do what the user asked, and nothing else?

THE IDEA

The AI picks from a catalog; it never writes code



Same pattern as **A2UI**, **CopilotKit** and **json-render**: they differ in packaging, not in the idea. The AI can only use what is in the catalog.

THE STACK

Why Vercel AI SDK + json-render

THE ENGINE

Vercel AI SDK

Talks to the model (any provider), streams the answer, and keeps the chat state in React. It has no generative-UI layer of its own.

THE GENERATIVE-UI LAYER

json-render (Vercel Labs)

Catalog → instructions, the UI JSON format, stream parsing and the renderer: the same pieces A2UI provides.

⚡ The patterns tested here are framework-independent: the findings carry over to A2UI or CopilotKit.

WHAT I BUILT

GenUI Studio: a chat that builds pages

19
components: 5 layout, 14 content

8
explicit rules for the agent

1
LLM call per message

11
automated test cases

Chat + live canvas
Preview, JSON view and version history (v1, v2...)

How it works page
Replays a real request in 13 steps with the actual data

Test suite
npm run eval scores how well the agent follows instructions

CASE 1

One agent, many page layouts

THE IDEA

Layout is just more components. Give the agent **Page, Stack, Grid, Columns** and **Card**, and one agent can build any page shape.

WHAT MADE IT WORK

Rules that say when to use each layout: several similar items → a grid; one main item plus details → two columns.

EVIDENCE

18 / 18

checks passed by the same agent

- Sales dashboard: 4-column metrics, chart, table
- Landing page: hero, 3 feature cards, button
- Sign-up form inside a card
- Two-column profile with a list

CASE 2

Components built from small components

THE IDEA

The AI returns a **tree**, not a picture: each element lists its children by id, so small pieces nest inside containers.

```
“A card titled Server Health with 3 progress bars and a warning alert”

Card “Server Health”
├ Progress CPU 84%
├ Progress Memory 62%
├ Progress Disk 91%
└ Alert warning: high disk latency
```

EVIDENCE

8 / 8

composition checks passed

Learned

The model sometimes used format features we didn’t plan for (state references). The renderer must support everything the format allows, or the rules must forbid it.

CASE 3

Following instructions: four controls

1 · Constrain

Only catalog components, with typed props (Zod)

2 · Instruct

Explicit rules: “exactly 3 means 3”, keep the order

3 · Edit with context

The current page goes with each follow-up; the agent patches only what changed

4 · Verify

3-layer validation on every page, and a test suite that scores it

EVIDENCE

38 / 38

checks passed with gemini-3.5-flash, in every case that ran

A lighter model added a banner nobody asked for: **model quality matters most.**

UNDER THE HOOD

One request, end to end

BROWSER

You type

useChat sends the whole conversation to /api/chat

→

SERVER

Adds context

The current page, plus instructions written from the catalog

→

GEMINI

One call

Writes a sentence, then JSON patch lines

→

SERVER

Sorts

pipeJsonRender splits text from UI patches and streams both

→

BROWSER

Builds and draws

Patches → page JSON → validation → React components

Everything streams, so the page grows while the model is still writing.

No tool calls: the model writes text in the format json-render teaches it.

EDITS

An edit changes only what was asked

YOU ASK

Make it dark with a green accent

THE MODEL RETURNS (REAL OUTPUT)

I will update the dashboard theme to dark mode with an emerald green accent.

```
{ "op": "replace",
"path": "/elements/page/props",
"value": { "accent": "emerald", "mode": "dark", … } }
```

1 patch

50 tokens out

Every other element unchanged